# Measurement

September 5, 2022

# Chocolate and Health

Read the short NYT article, "In One Study, a Heart Benefit for Chocolate" in "Readings" folder on CANVAS.

Then, evaluate whether the study establishes *causation*:

1. There is covariation between **X** and **Y**
2. There is a plausible causal mechanism
3. **X** causes **Y**, not the other way around (temporal order)
4. Confounders are ruled out or controlled for

# Observational Studies

The researcher makes comparisons of units that already exist in the real world.

- Researcher does not have control over independent variable assignment
- *Approximate* experiment by explicitly accounting for confounders
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Can we establish causation with observational studies?

# Evaluating Research Designs

1. To what extent does it establish a causal relationship?

2. Internal Validity- Are we confident in our conclusions with respect to the specific subjects being studied?

3. External Validity- Can we generalize our conclusions to the broader population?
   - External sample validity- does our sample represent the broader population?
   - External validity of stimulus- is the treatment capturing the effect it would actually pick up in the real world?

# Evaluating Research Designs

*In general*, experiments are high in internal validity but low in external validity.

But both types of studies can face threats to both types of validity.

- Threats to Internal Validity (a few examples)
  - Weird issues that pop up between measurements
  - Hawthorne Effect
  - Mortality
- Threats to External Validity (a few examples)
  - Poor representation in sampling
  - Reactivity

Questions on Research Design?

# Measurement

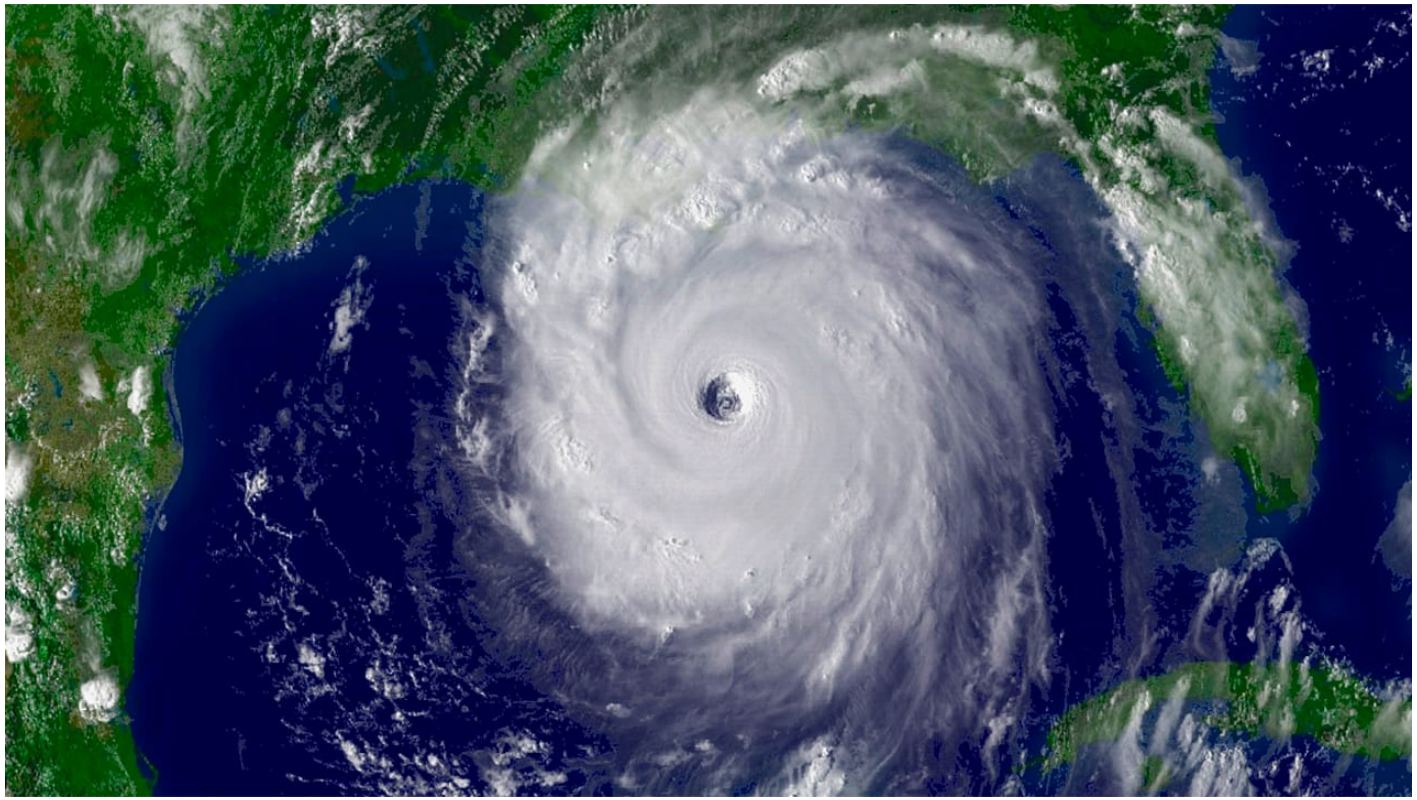
Color in all states in the South.

# Measurement

Defining observable units of abstract concepts

- Remember, theories explain *relationships* between *concepts*
- Testing theories empirically means we must measure those concepts
- Then, we can observe patterns systematically
- And, of course, establish relationships between variables

# Hurricane

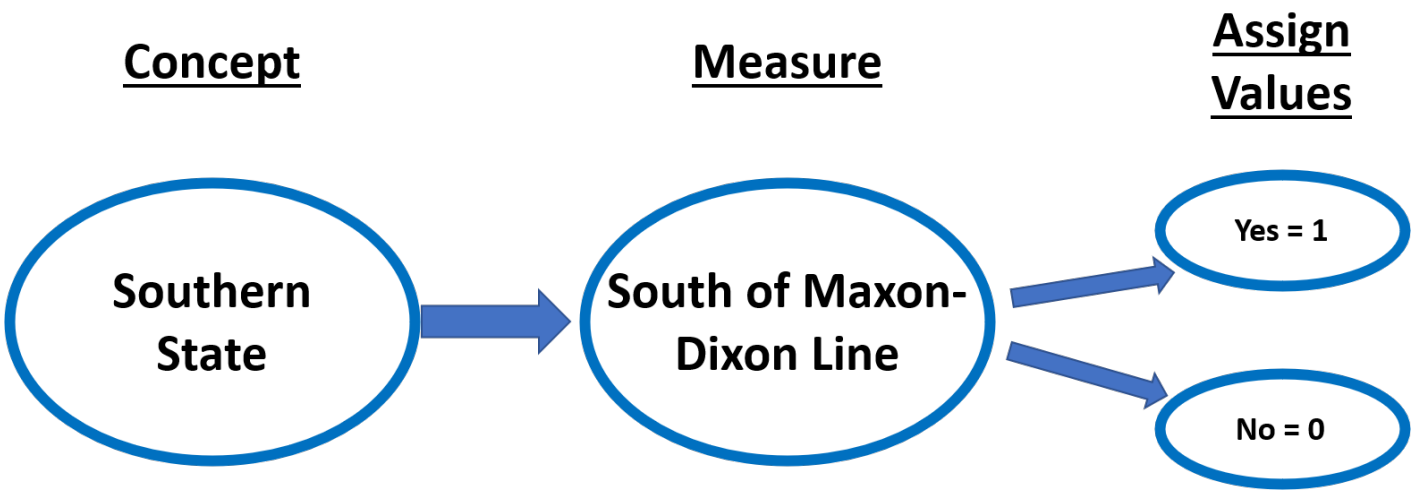


# Planets

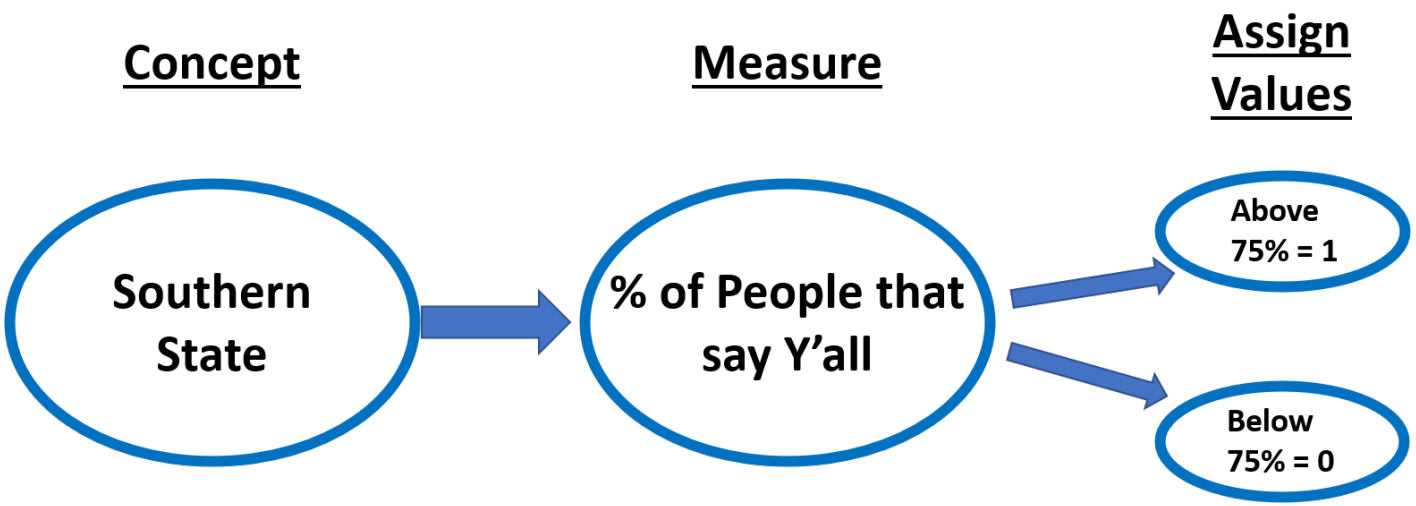
NASA: Since there are no more planets to be discovered in our solar system, we are expanding our search further.
Pluto:

# “States”

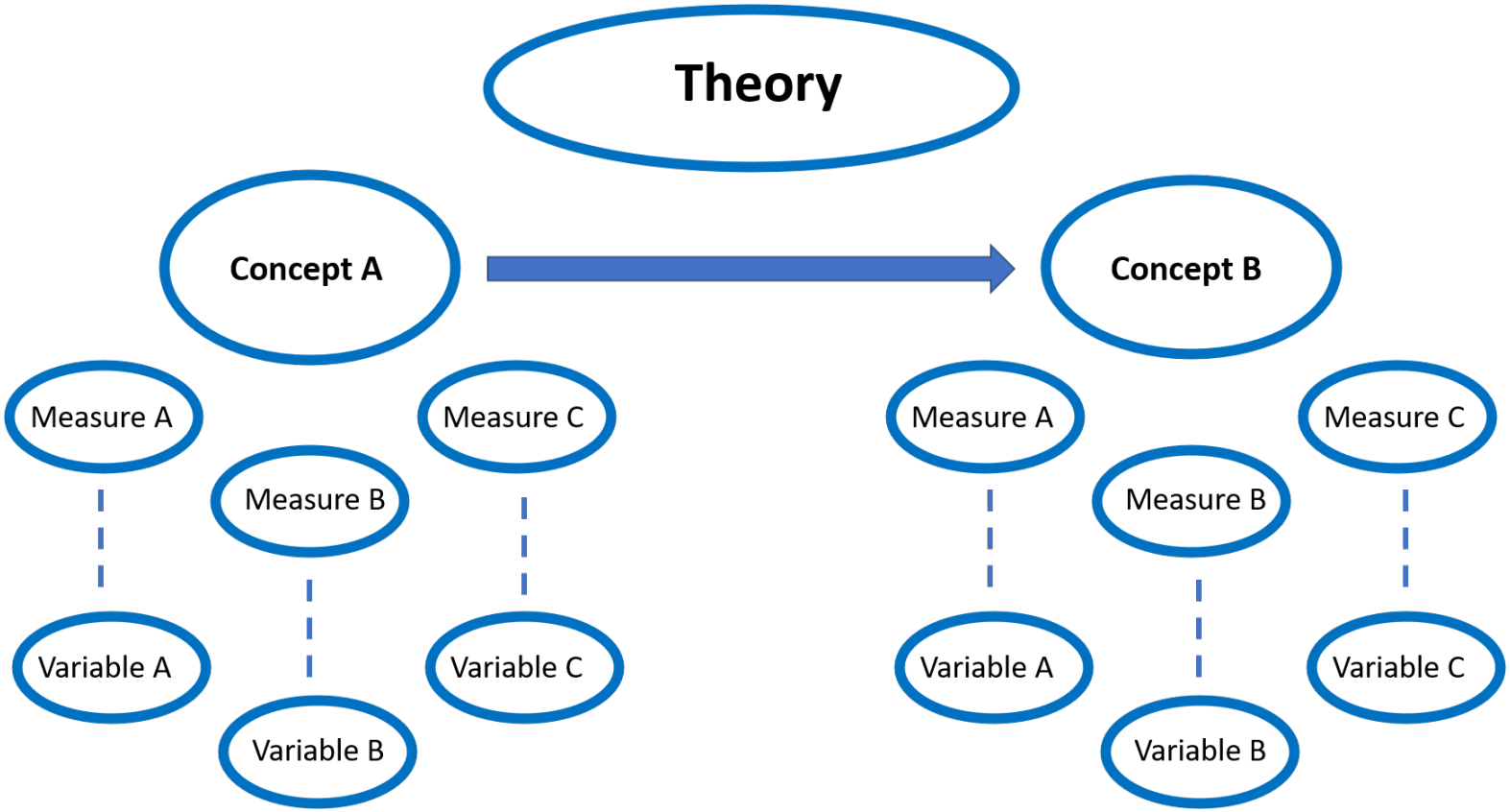
# Measuring the South I



# Measuring the South II



# Multiple Possible Measurement Strategies



# Measuring Democracy

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Democracy-Dictatorship Measure (Cheibub, Gandhi, and Vreeland 2010): Democracy or Dictatorship

1. The chief executive is elected
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Polity Score: Ranges from -10 (Super Autocratic) to 10 (Super Democratic)

- Competitiveness of executive recruitment, openness of executive recruitment, constraints on executive, regulation of political participation, competitiveness of political participation.

# Importance of Measurement

Why is careful measurement important?

- Case selection- studying the relevant subjects
- Accuracy of variables- studying the variation/relationships we are interested in
- Meaningful Tests of Theory- evaluating theory properly

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Three main types of validity:

1. Face Validity- Is the measure reasonable? Does it make sense at face value? (subjective, initial examination)
   - Star Wars Fan (Don't measure by asking if they like Star Trek)
2. Construct Validity- Does the measure pick up the important related elements of the concept?
   - Diagnosing Depression (Questionnaire for symptoms, but is it actually picking up mood or self-esteem?)
3. Content Validity- Is the measure fully representative of the elements that comprise the concept?
   - Democracy (Capture all the elements of democracy?)

# Evaluating Our Measures II

Reliability- The tendency of our measure to consistently produce the same value for the same observation

- Applying the same measurement rules to the same observation will produce identical results
- Unreliable measures produce inconsistent results for the same observation

# Measurement Error

Systematic Error- Some factor consistently affects measurement

- Leads to **bias** in values
- i.e., consistently more positive values than should be or consistently more negative values than should be
- Very bad if we cannot identify the source and account for it
- Example: Studying and exams

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Random Error- Some factor may make measurement inaccurate for some cases, but not consistently positive or negative because source is random itself (a.k.a. "noise")

- Does not affect averages
- Example: Distribution of moods affecting performance on exam

# Validity and Reliability

Unreliable & Unvalid

Unreliable, But Valid

Reliable, Not Valid

Both Reliable & Valid

# Eck and Fariss (2018)

Sweden and Human Rights Abuse: Why is it hard to measure human rights violations with reports?

# Project Stuff

# Next Class

- Intro to R:
  1. Read: None
  2. Follow slides on CANVAS (posted soon) to install R and RStudio (in that order)
- Be thinking about a research topic! Come see me!
- Come to class with your computer on Tuesday!